

🔪 Module A: Probability basics (expectation, variance, independence)

Big picture Data in machine learning are random, so we describe them with **distributions** and summarize them with **expectations** (averages) and **variances** (spread).

Distributions. Discrete: $p(x) \geq 0$ and $\sum_x p(x) = 1$. Continuous: a density $p(x) \geq 0$ with $\int p(x) dx = 1$.

Expectation. $\mathbb{E}[f] = \sum_x p(x) f(x)$, or $\int p(x) f(x) dx$. It is linear: $\mathbb{E}[af + bg] = a\mathbb{E}[f] + b\mathbb{E}[g]$, and $\mathbb{E}[c] = c$ for a constant c .

Variance & covariance. $\text{var}[f] = \mathbb{E}[(f - \mathbb{E}[f])^2]$ and $\text{cov}[x, y] = \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])]$.

Independence. x and y are independent if $p(x, y) = p(x)p(y)$.

Three distributions to know.

Bernoulli: $p(x | \mu) = \mu^x (1 - \mu)^{1-x}$ for $x \in \{0, 1\}$.

Uniform: $U(x | a, b) = \frac{1}{b-a}$ for $a \leq x \leq b$.

Gaussian: $\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

⚠️ Remarks

- $\mathbb{E}[x + y] = \mathbb{E}[x] + \mathbb{E}[y]$ **always** holds, but $\mathbb{E}[xy] = \mathbb{E}[x]\mathbb{E}[y]$ **needs independence**.
- $\mathbb{E}[f]$ is just a number, so it comes out of another expectation: $\mathbb{E}[f \cdot \mathbb{E}[f]] = \mathbb{E}[f] \cdot \mathbb{E}[f]$.

👉 Practice: Bishop 1.5, 1.6 & 1.10 (variance and independence)

(a) Starting from the definition $\text{var}[f] = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2]$, show that $\text{var}[f] = \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2$. *Hint:* expand the square and use linearity; $\mathbb{E}[f(x)]$ is a constant.

(b) Show that if two variables x and y are independent, then their covariance is zero. *Hint:* as in (a), first show that $\text{cov}[x, y] = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$.

(c) Suppose that x and z are independent. Show that $\mathbb{E}[x + z] = \mathbb{E}[x] + \mathbb{E}[z]$ and $\text{var}[x + z] = \text{var}[x] + \text{var}[z]$.

👉 Practice: Bishop 2.1 & 2.12 (Bernoulli and uniform)

(a) Verify that the Bernoulli distribution $p(x | \mu) = \mu^x (1 - \mu)^{1-x}$, $x \in \{0, 1\}$, satisfies $\sum_{x=0}^1 p(x | \mu) = 1$, $\mathbb{E}[x] = \mu$ and $\text{var}[x] = \mu(1 - \mu)$.

(b) The uniform distribution is $U(x | a, b) = \frac{1}{b-a}$ for $a \leq x \leq b$. Verify that it is normalized, and find its mean and variance.

📌 Module B: Linear algebra & calculus basics (vectors, matrices, derivatives)

Big picture Data are stored as **vectors and matrices**, and a model is trained by taking **derivatives** and setting them to zero.

Dot product & norm. For $a, b \in \mathbb{R}^n$: $a^\top b = \sum_i a_i b_i = b^\top a$ and $\|a\|^2 = a^\top a$. Expanding,

$$\|a - b\|^2 = \|a\|^2 - 2a^\top b + \|b\|^2.$$

Transpose & inverse. $(AB)^\top = B^\top A^\top$. For invertible $n \times n$ matrices A, B : $(AB)^{-1} = B^{-1}A^{-1}$ and $(A^{-1})^\top = (A^\top)^{-1}$.

Symmetric: $A^\top = A$. **Antisymmetric:** $A^\top = -A$. A number (1×1) equals its own transpose, e.g. $x^\top Ax = (x^\top Ax)^\top = x^\top A^\top x$.

Chain rule. $(f(g(a)))' = f'(g(a)) g'(a)$.

Logistic sigmoid. $\sigma(a) = \frac{1}{1 + e^{-a}}$ squashes any real score into $(0, 1)$, so it can be read as a probability (logistic regression, week 2).

Minimize by setting the derivative to 0. E.g. $\frac{d}{dc} \sum_n (x_n - c)^2 = -2 \sum_n (x_n - c) = 0$ gives $c = \frac{1}{N} \sum_n x_n$: the mean is the best constant fit in squared error (least squares, week 1).

⚠️ Remarks

- Transposes **reverse the order**: $(AB)^\top = B^\top A^\top$, not $A^\top B^\top$.
- Chain rule: don't drop the inner derivative, e.g. $\frac{d}{da} e^{-a} = -e^{-a}$.

🔪 Practice: Bishop 2.22 & 2.17 (symmetric matrices)

- (a) Show that the inverse of a symmetric matrix is itself symmetric.
- (b) Any square matrix splits as $\Lambda = \Lambda_S + \Lambda_A$ with $\Lambda_S = \frac{1}{2}(\Lambda + \Lambda^\top)$ symmetric and $\Lambda_A = \frac{1}{2}(\Lambda - \Lambda^\top)$ antisymmetric. Show that $x^\top \Lambda_A x = 0$ for every vector x , so only the symmetric part appears in $x^\top \Lambda x$. (This is why a Gaussian's covariance matrix can be taken symmetric.)

🔪 Practice: ESL 2.1 (one-hot targets)

Suppose each of K classes has an associated target $t_k \in \mathbb{R}^K$, a vector of all zeros except a one in the k th position. Show that classifying to the largest element of $\hat{y} \in \mathbb{R}^K$ amounts to choosing the closest target, $\min_k \|t_k - \hat{y}\|$, if the elements of $\hat{y}$ sum to one. *Hint:* expand $\|t_k - \hat{y}\|^2$ using the review box.

🔪 Practice: Bishop 4.7 & 4.12 (the logistic sigmoid)

For $\sigma(a) = \frac{1}{1 + e^{-a}}$: (a) show that $\sigma(-a) = 1 - \sigma(a)$;

(b) show that its inverse is $\sigma^{-1}(y) = \ln \frac{y}{1 - y}$;

(c) verify that $\frac{d\sigma}{da} = \sigma(a) (1 - \sigma(a))$.